



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Assignment-01 || Summer 2024

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

Total Marks: 10

Deadline: 20.09.24

Answer all questions. Figures in the right-hand margin indicates full marks.

1. Design DFAs that accepts the following languages:

1 x 4

- a) $L = \{ w \mid w \text{ starts and ends with } \mathbf{different} \text{ symbol, having } \mathbf{even} \text{ length strings when starts with } \mathbf{a} \text{ or } \mathbf{odd} \text{ length strings when starts with } \mathbf{b} \}$ where, $\Sigma = \{a,b\}$
- b) $L = \{ w \mid w \text{ starts with "ab", contains "cca" as a substring and ends with "aba"} \}$ where, $\Sigma = \{a,b,c\}$
- c) $L = \{ w \mid w \mathbf{neither}$ contains two consecutive a's $\mathbf{nor}$ two consecutive b's $\}$ where $\Sigma = \{a,b\}$
- d) $L = \{ w \mid w \text{ only accepts binary strings those are a } \mathbf{multiple} \text{ of 4.} \}$
 $\mid \Sigma = \{0, 1\}$

2. Design NFAs that accepts the following languages:

1 x 4

- a) $L = \{ w \mid w \text{ starts with "12" or "32" and contains "222" or "111" or "333" and ends with "31" or "23"} \mid \Sigma = \{1,2,3\}$
- b) $L = \{ w \mid w \text{ double "1" is followed by triple "0" and starts with "0"} \mid \Sigma = \{0,1\}$
- c) $L = \{ w \mid w \text{ third symbol from the right end is always "a" and the second symbol is either "b" or "c"} \mid \Sigma = \{a,b,c\}$
- d) $L = \text{contains 'xx' or 'yx' or 'zz' and ends with 'yz' or 'y' } \mid \Sigma = \{x,y,z\}$

3. Consider the following NFA, and show with the help of NFA-tree whether the string **aabbbbbaa** is accepted or not.

2

